

Quantum-Resistant Architecture Begins at the Chip

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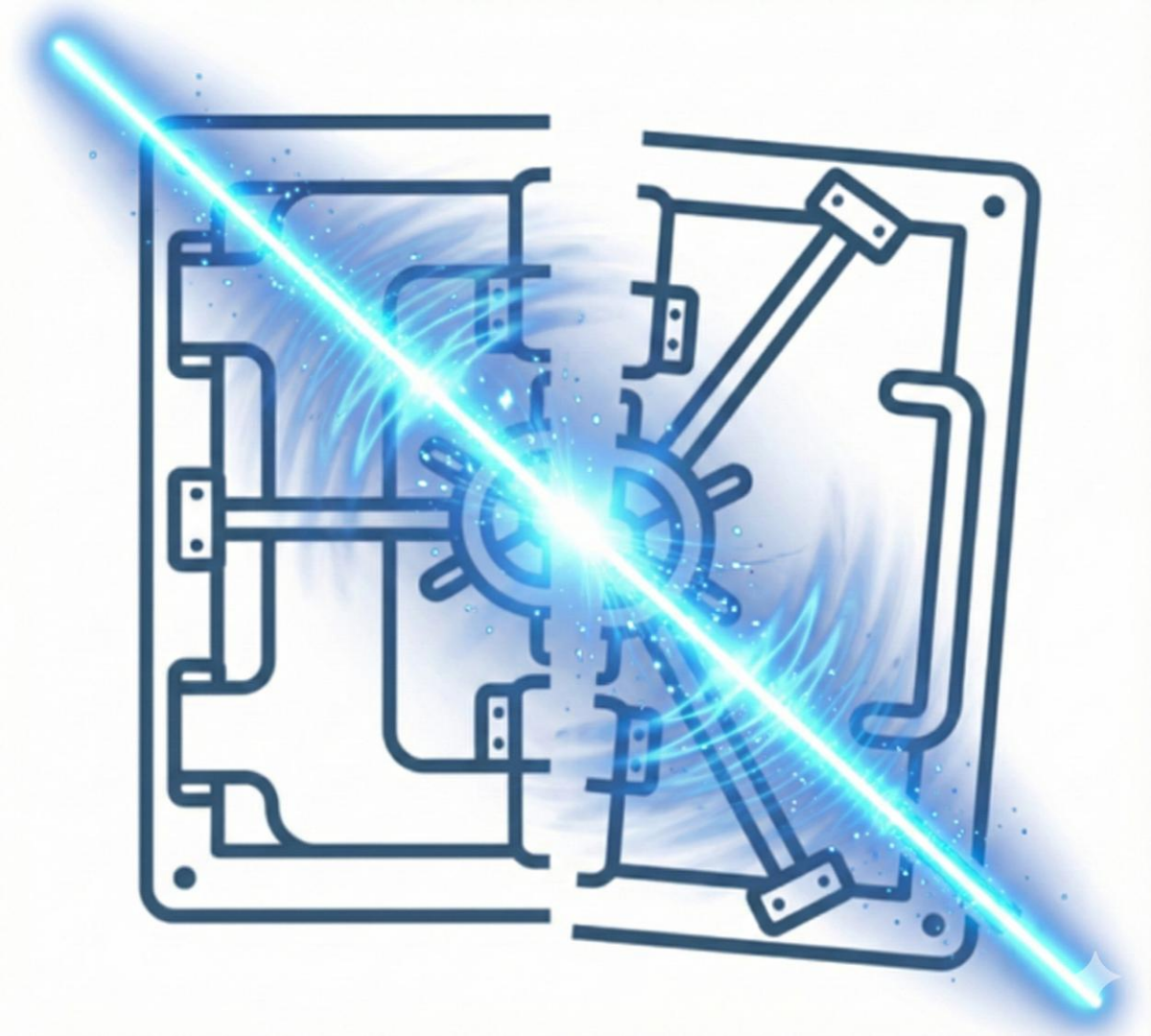
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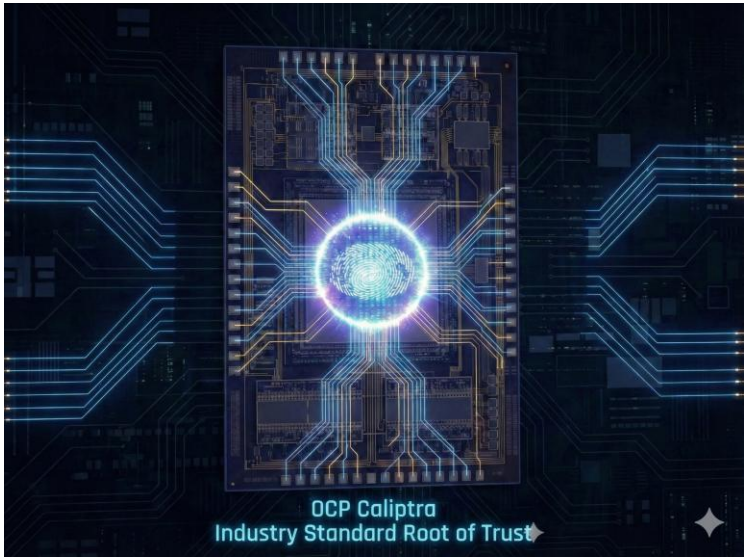
Mathematics-based security is over.

Quantum computers break it in seconds.

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Quantum Resilience with CNSA 2.0 & Caliptra

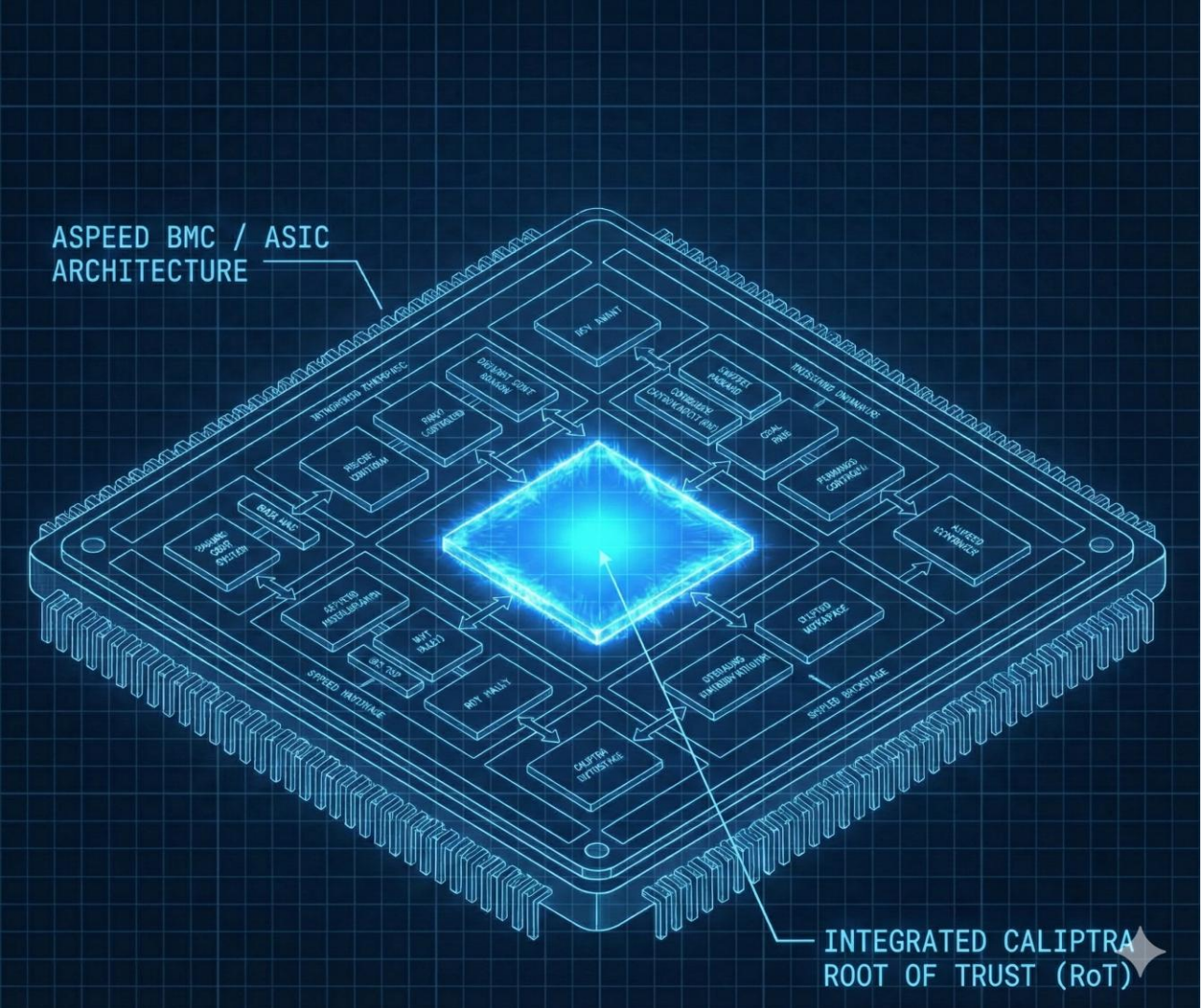




Mandatory cryptographic migration.

Embedding post-quantum security directly into the silicon to create an immutable Root of Trust.

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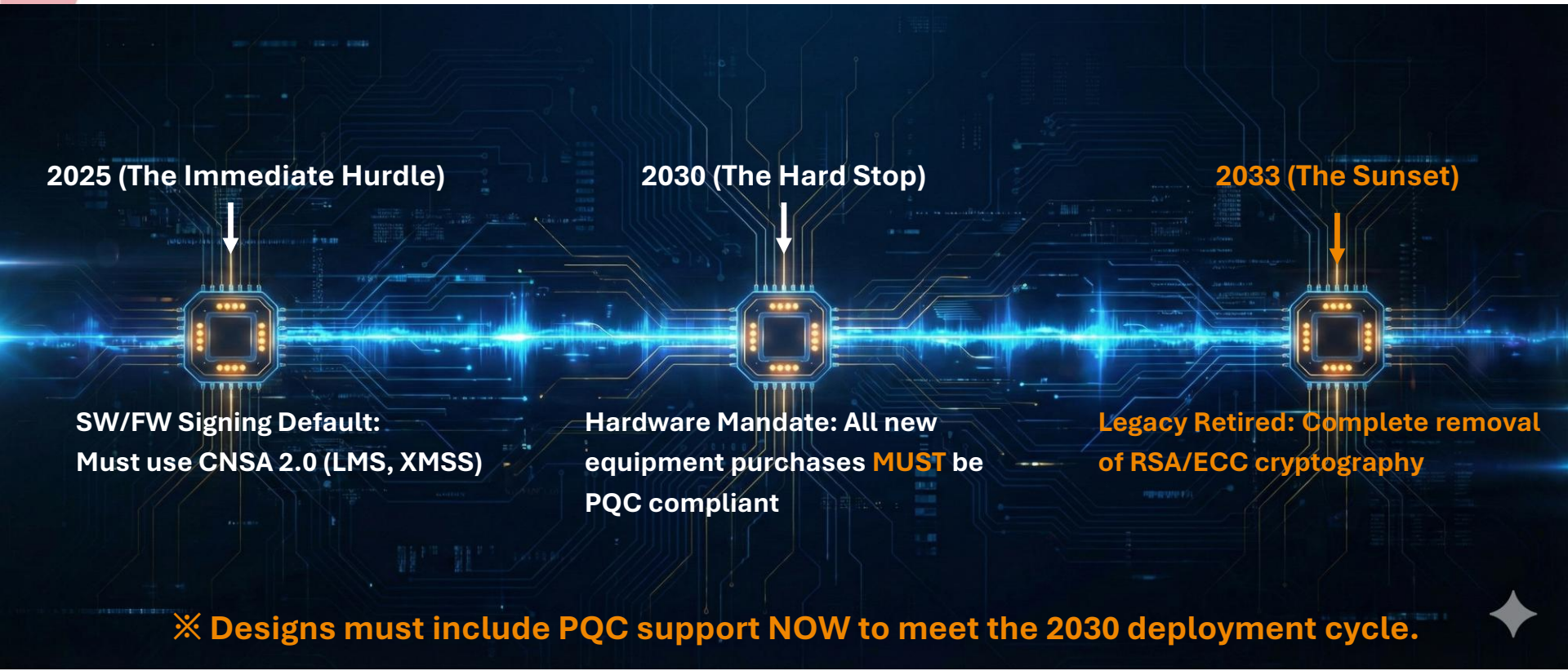
NSA CNSA 2.0 Compliant

*"**ASPEED** is the first adopter of **OCP Caliptra**, embedding post-quantum security directly into the silicon to create an immutable **Root of Trust**"*

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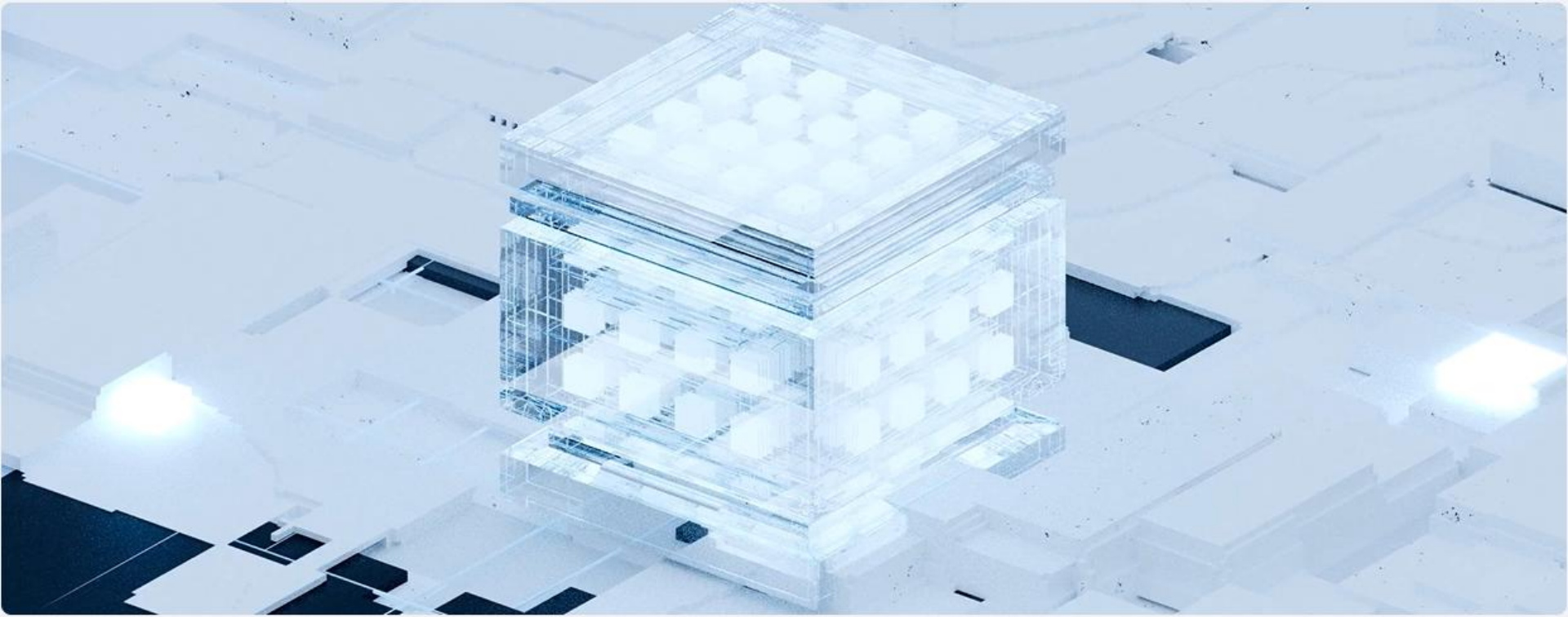
The CNSA Migration Timeline

Aggressive. Non-Negotiable. Mandatory.



※ Designs must include PQC support NOW to meet the 2030 deployment cycle.

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Caliptra is supported by industry leaders.

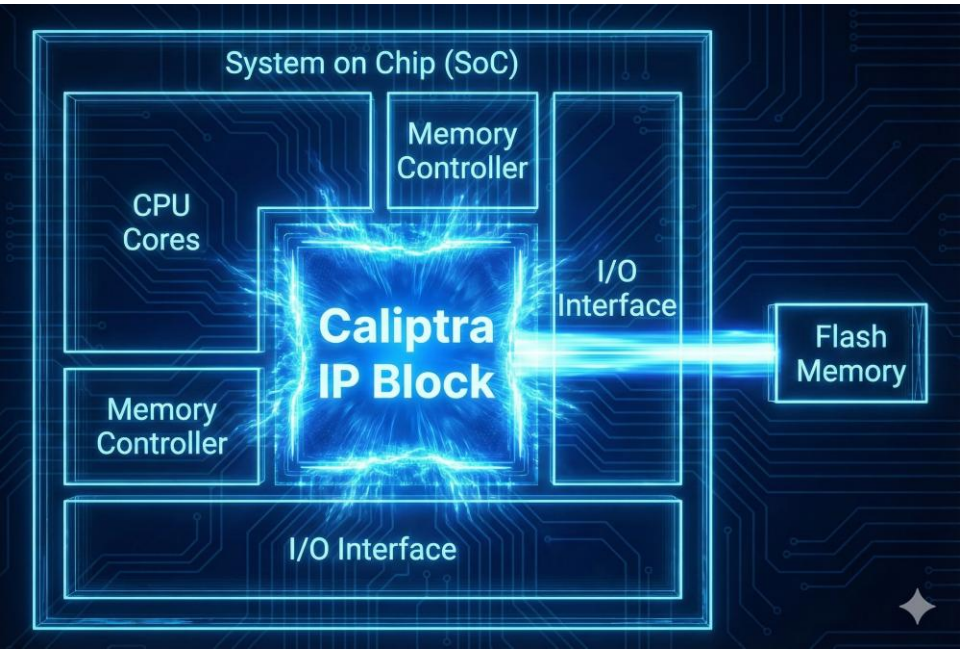
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Secure Foundation for BMC, CPU, GPU, DPU, and TPU

A reusable Hardware IP block embedded directly into the silicon.



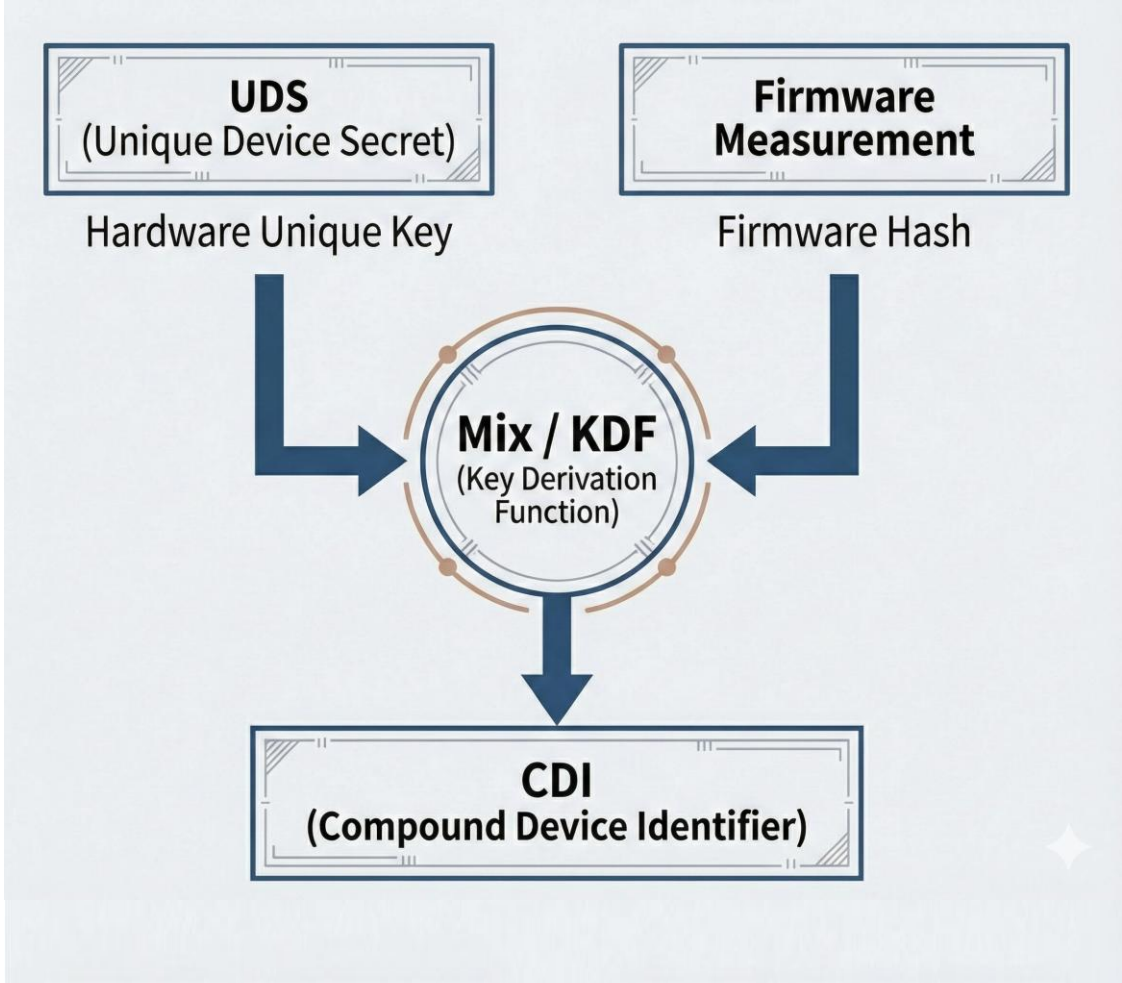
- *Root of Trust for Measurement (RTM)*
- *Secure Boot*
- *PQC algorithms*
- *Caliptra consists of:*
  - ◎ RTL (hardware)
  - ◎ ROM code
  - ◎ Firmware code

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Identity Origin

DICE – Device Identifier Composition Engine

- *UDS is the hardware unique key, accessible only by Caliptra.*
- *Measurement: hash operations*



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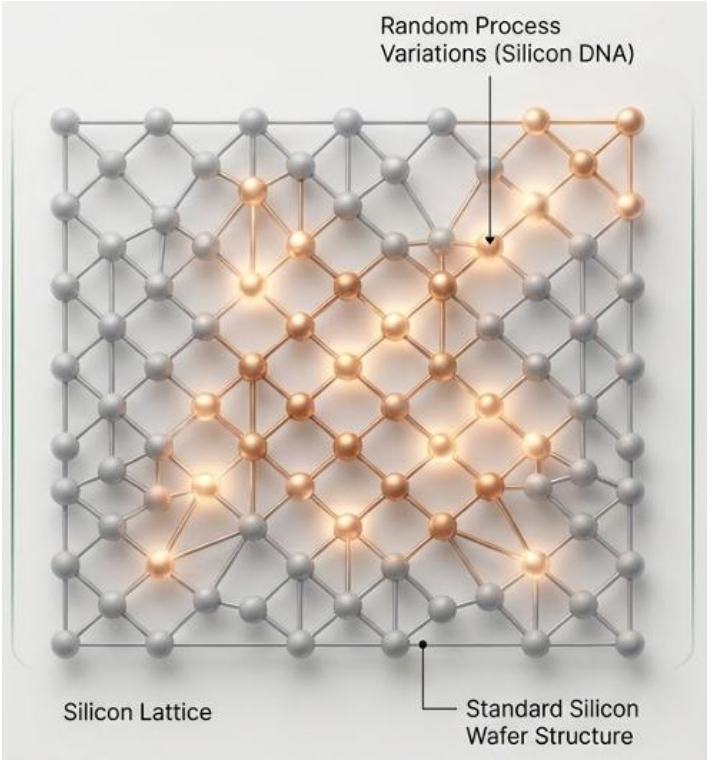
Securing Caliptra's Foundation with PUF

Meets Caliptra's requirements for a high-entropy, unique secret.

PUF: Superior Source for the Unique Device Secret (UDS) for Caliptra.

"Even the manufacturer cannot create two identical chips."

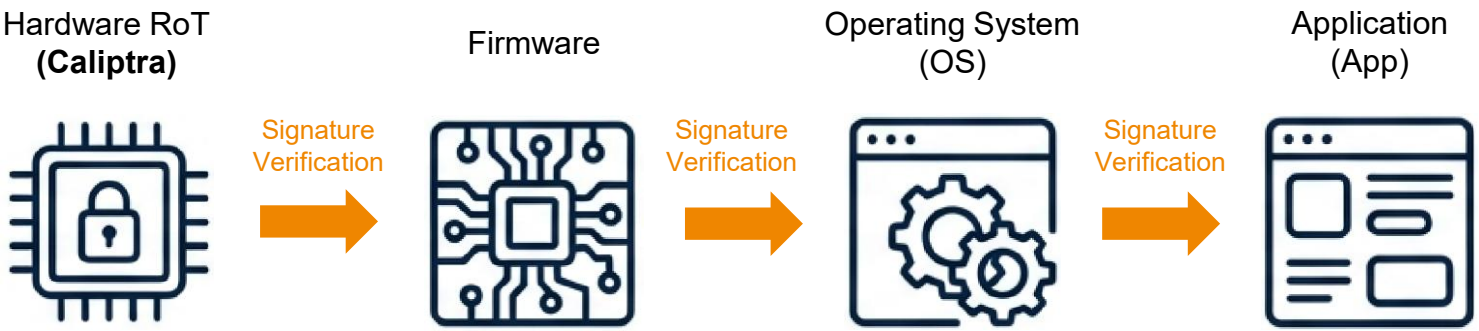
High entropy *powered by PUF is perfect for* ***quantum-resistant*** *architecture.*



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How ASPEED Build Total Integrity

Blocking threats with Chain of Trust.



Secure Boot

Verify first, then execute.

Block low-level malware.

Silicon Root of Trust

Hardware cannot be tampered.

Establish a chain of trust.

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OCP S.A.F.E Compliance

Standardizing security audits for the supply chain.

OPEN Compute Project SOLUTION PROVIDER®

OCP S.A.F.E.

The Problem: Traditional security audits are fragmented.

The Solution: A standardized framework.

3rd Party Verification: Independent security labs approved by OCP.

Supply Chain Trust: Transparent, objective proof that ASPEED's firmware and hardware meet rigorous security standards.

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PQC is Evolving: Future-Proofing Our Solutions

Embracing the dynamic nature of post-quantum cryptography.

The Standard in NOT Static.

Crypto-Agility is Key.

Continuous Monitoring.

Long-Term Commitment.

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Fusing PUF, PQC, and Caliptra to Build Quantum-Resistant Data Cener



Caliptra (Standard)

This industry standard Root of Trust is PQC-ready and execute boot measurements.

PUF (Entropy)

True randomness derived from physical silicon variations (the silicon “fingerprint”),

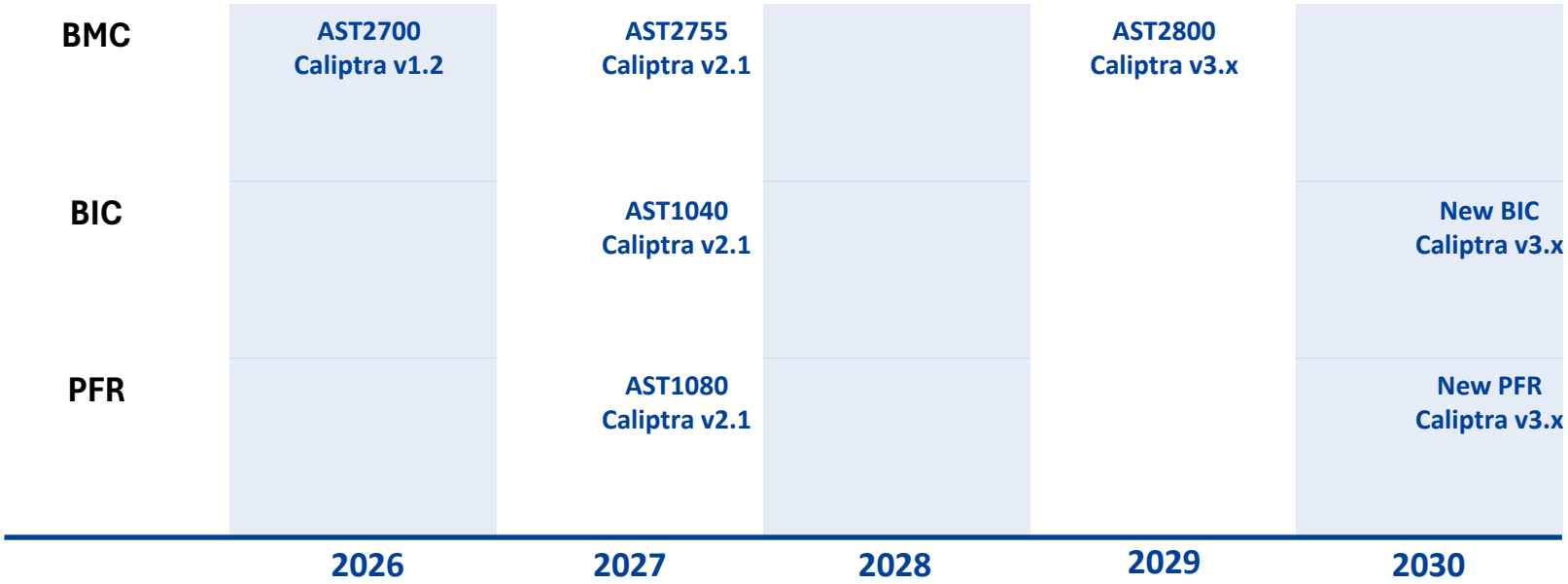
PQC (Algorithm)

NIST-standard algorithms designed to resist quantum attacks, and it requires high-quality entropy.

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Roadmap to Quantum-Resilience

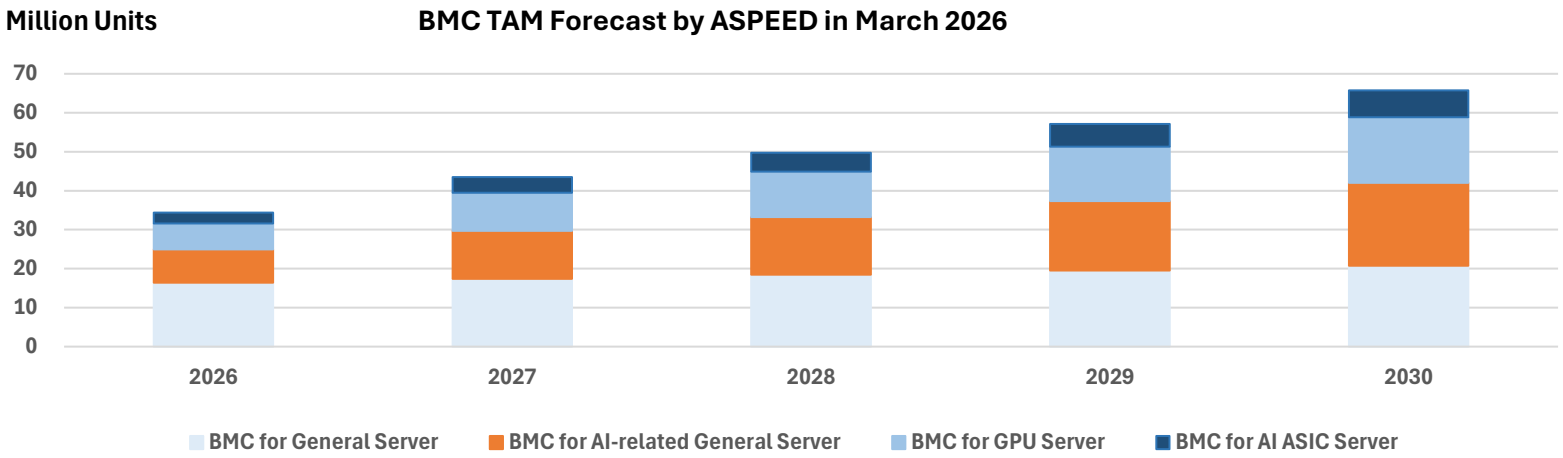
Delivering quantum-resilient ***OCP Caliptra security*** across the entire ***ASPEED*** product portfolio.



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BMC TAM Forecast Revised

- We see huge demand for **AI-related General Server** for agentic AI workload and simple inference tasks.
- BMC TAM is estimated to be **65.77 million** units in 2030.



Note：1) BMC for General Servers will grow at 6% CAGR.
2) BMC for AI-related General Servers is estimated to be **8.5 million** in 2026, with 45% growth rate in 2027 and 20% thereafter.
3) BMC for GPU servers and AI ASIC servers in 2026 is based on the estimate of GPU and AI ASIC shipment and the server architecture.
4) Assuming BMC for GPU Servers grow at 45% in 2027 and 20% afterwards. We also include BMCs for ICMS here.
5) Assuming BMC for AI ASIC Servers grow at 45% in 2027 and 20% afterwards.

Source：ASPEED estimate

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2026 Q2 Guidance

For 2026 Q2, we currently expect:

- Consolidated Revenue: Around **NT$3.4 billion ~ 3.6 billion**
- Gross margin: **67% ~ 68%**

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